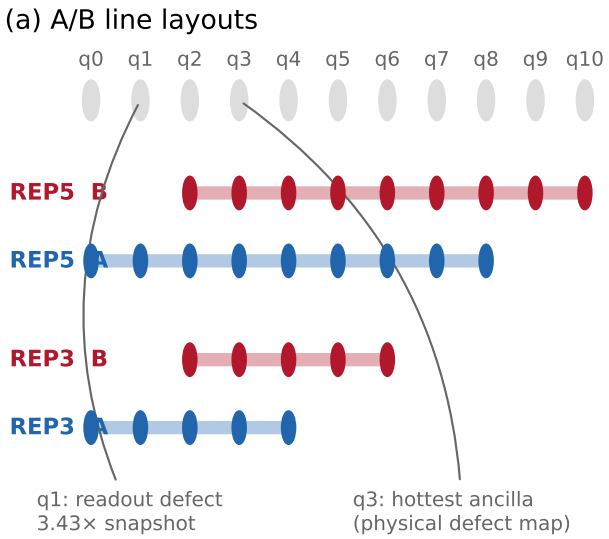
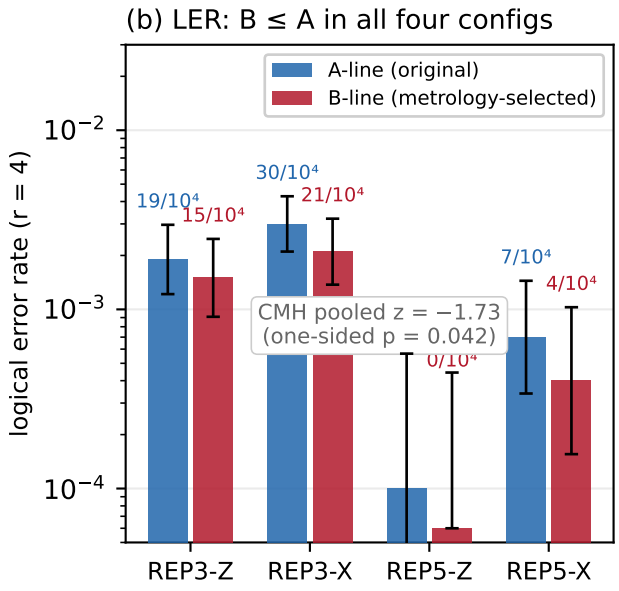
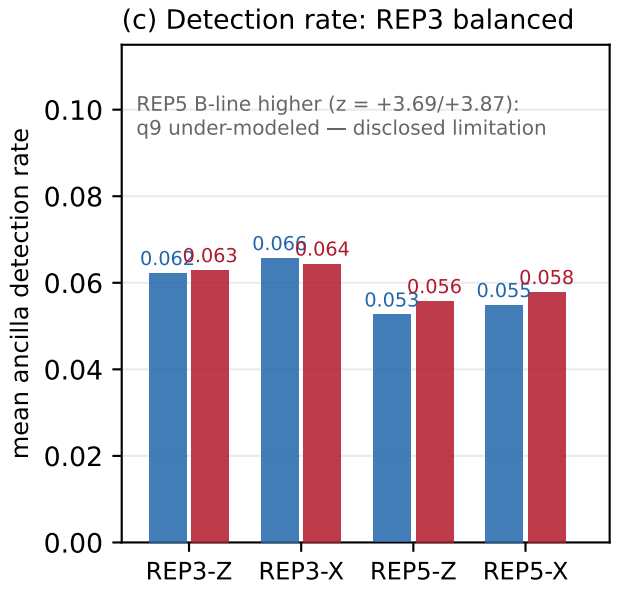

(a) A/B line layouts
q0 q1 q2 q3 q4 q5 q6 q7 q8 q9 q10
REP5 B
REP5 A
REP3 B
REP3 A
q1: readout defect
3.43× snapshot
q3: hottest ancilla
(physical defect map)


(b) LER: B ≤ A in all four configs
A-line (original)
B-line (metrology-selected)
logical error rate (r = 4)
10^{-2}
10^{-3}
10^{-4}
19/10⁴
15/10⁴
30/10⁴
21/10⁴
0/10⁴
7/10⁴
4/10⁴
CMH pooled z = −1.73
(one-sided p = 0.042)
REP3-Z
REP3-X
REP5-Z
REP5-X


(c) Detection rate: REP3 balanced
REP5 B-line higher (z = +3.69/+3.87):
q9 under-modeled — disclosed limitation
mean ancilla detection rate
0.10
0.08
0.06
0.04
0.02
0.00
0.062
0.063
0.066
0.064
0.053
0.056
0.055
0.058
REP3-Z
REP3-X
REP5-Z
REP5-X